



Regen Stride
RESILIENCE, WOVEN INTO EVERY STEP

INDUSTRY: Athletic Footwear

FOUNDED: 2026

LOCATION: Ithaca, NY

FUNDRAISING:

Pre-Seed Stage: Currently seeking \$750K–\$1M. Use of funds focused on polymer optimization, third-party validation, and OEM pilot integration

VERTICAL: Athletic Footwear Materials Technology

PATENTS/IP: US Patent 10,745,509 B2: Methods of Making Mechanoresponsive Healing Polymers Gordon & Siochi · NASA Filed Jul 11, 2014

EXECUTIVE TEAM:

Aislinn Ennis | Chief Executive Officer
508-769-2769 | ace68@cornell.edu

Anthony Bartolotta | Chief Operating Officer
516-761-7188 | asb374@cornell.edu

Aryan Baweja | Chief Revenue Officer
(437) 972-7926 | ab3383@cornell.edu

Chloe Teig | Chief Strategy Officer
(646) 763-0189 | crt78@cornell.edu

Kavya Mittha | Chief Market Officer
(408) 838-3676 | km773@cornell.edu

Riya Guttigoli | Chief Financial Officer
(469) 994-6557 | rag323@cornell.edu

TAGLINE: Resilience, woven into every step.

SUMMARY: RegenStride is commercializing a licensed self-healing polymer IP by integrating it into athletic footwear midsoles to extend useful shoe life beyond the typical 400-mile cushioning breakdown point. We license the technology to a single elite OEM partner, earning recurring royalties as the material expands from one launch model into broader product lines.

PROBLEM: After about 400 miles, runners discard their sneakers because the midsole loses its cushioning and protective performance, long before the rest of the shoe wears out. Our wide customer interview range included 51 people including many runners and OEM engineers, and they confirmed this as the primary failure point. Market forces are driving urgency because the global athletic footwear market is growing at over 5% CAGR, consumers are increasingly sustainability-conscious and resistant to premature replacement cycles, and OEMs are actively seeking materials differentiation as competition intensifies. Current premium footwear technologies improve energy return, speed, or outsole durability, but do not position retained cushioning as the central value proposition, creating a clear window of opportunity for a novel material solution.

SOLUTION: RegenStride integrates a self-healing polymer into footwear midsoles to improve retained cushioning and extend usable lifespan. Unlike existing performance foams or plated trainers, RegenStride directly targets long-term cushioning retention. Our commercialization strategy begins with a single strategic OEM partner, initially targeting Nike due to its R&D infrastructure and proven limited-edition launch ecosystem.

TARGET MARKET: The global athletic footwear market represents an estimated \$131B TAM, with Nike footwear alone representing an estimated \$51B SAM. RegenStride's SOM lies within the projected five-year licensing opportunity including a single OEM partnership is approximately \$17.5M annually. Target customers include footwear innovation teams, materials engineers, and performance footwear product groups at major athletic brands.

COMPETITIVE ADVANTAGE: Current footwear technologies improve energy return or outsole durability but do not directly address retained cushioning as a core value proposition. RegenStride differentiates itself through self-healing cushioning retention, OEM differentiation potential, sustainability benefits through reduced replacement frequency, and scalable licensing economics.

BUSINESS MODEL: RegenStride operates as a B2B licensing company, earning an estimated 8–10% royalty on wholesale footwear price, or approximately \$5.60–\$7.00 per pair sold. Our commercialization pathway begins with internal validation testing and a strategic OEM partnership, followed by a limited “halo” launch, standard product integration, and eventual multi-SKU scaling across footwear categories. Projected Year 5 annual revenue is approximately \$14M–\$17.5M, with estimated customer economics of \$80K–\$150K CAC and an LTV:CAC ratio of approximately 1:7.5–1:10. Pre-seed funding will support polymer optimization, third-party testing, OEM pilot validation, and commercial partnership development.

BUSINESS MODEL CANVAS

Key Partners

- Exclusive OEM partner (integration & revenue share)
- Licensing / manufacturing partner (royalties)
- Validation labs (spec-in)

Key Activities

- Polymer formulation & optimization
- Impact/fatigue testing under cyclic loading
- Integration into midsole materials
- Prototype footwear testing

Key Resources

- Self-healing polymer IP
- Polymer formulation for elastomers/foam
- Cyclic fatigue & compression testing

Value Proposition

- Reduces midsole material and structural material fatigue
- Maintains cushioning and energy return under loading
- Extends usable lifespan of performance footwear
- Enables premium long-lasting running shoes

Cost Structure

- R&D & testing
- Raw polymer inputs
- Manufacturing scale-up
- Prototype footwear integration
- IP protection

Customer Relationships

- Direct OEM engineer engagement & evaluation
- Support through sampling, testing, validation
- Spec-in → expand & long-term agreements

Channels

- Direct virtual outreach to OEM design engineers
- Prototype validation
- Direct commercial path to OEMs
- Licensing pathway through strategic partner
- Customer Relationships
- Direct OEM engineer engagement & evaluation
- Support through sampling, testing, validation
- Spec-in → expand & long-term agreements

Customer Segments

- Athletic footwear materials engineers
- Footwear product development teams at athletic brands
- Innovation/advanced materials teams at sporting shoe brands

Revenue Streams

- B2B Licensing (% of OEM sales)
- Pilot / prototype fees
- DTC Production revenue share (phase 2)

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